EV Charger Simulator

Validate OCPP charging operations without physical hardware

What it is

The EV Charger Simulator is a production-focused **OCPP 1.6J simulation toolkit** that behaves like real charging stations over WebSocket. Teams use it to test onboarding, charging sessions, billing logic, and remote command handling before deploying to live infrastructure.

Why teams use it

â¢ **Launch faster:** test charger workflows on day one, without waiting for devices in the field.
â¢ **Lower QA cost:** run repeatable automated scenarios instead of manual hardware test cycles.
â¢ **Catch edge cases early:** validate resets, reconnects, meter bursts, and fleet surges in controlled environments.
â¢ **Scale confidently:** simulate dozens of connectors and multiple stations in parallel.

Core capabilities

â¢ Full charger lifecycle support:
- ‘BootNotification‘, ‘StatusNotification‘, ‘Heartbeat‘
- ‘Authorize‘, ‘StartTransaction‘, ‘MeterValues‘, ‘StopTransaction‘
â¢ Handles server-initiated commands, including:
- ‘RemoteStartTransaction‘, ‘RemoteStopTransaction‘, ‘Reset‘
- ‘ChangeConfiguration‘, ‘GetConfiguration‘
- ‘SetChargingProfile‘, ‘ClearCache‘, ‘UnlockConnector‘
â¢ Meter values for realistic downstream testing:
- Energy, power, current, and voltage streams

Four usage modes

â¢ **boot** â keeps a simulated charger online for manual command testing.
â¢ **session** â executes one complete charge session flow.
â¢ **multi** â runs concurrent sessions across all connectors.
â¢ **fleet** â launches many chargers simultaneously for load and stress tests.

Built for engineering and operations

â¢ Supports local development, CI pipelines, and staging validation.
â¢ Accelerated time mode can compress long sessions into short test windows.
â¢ Helps verify platform behavior before touching customer-facing systems.

Typical outcomes

â¢ Faster integration with OCPP-compliant backends
â¢ Better reliability of remote operations
â¢ More predictable releases with fewer field regressions
â¢ Improved confidence in billing and transaction analytics

Quick start

```
npm install
cp config.example.json config.json
npm run dev boot
```

Best fit

The simulator is ideal for EV charging platforms, CPO engineering teams, integrators, and QA organizations that need dependable, repeatable charger behavior at scale.